API SECURITY RISK ANALYSIS REPORT

Cyber Security Internship – Future Interns

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1. Introduction

This report presents an API Security Risk Analysis conducted using Postman on public APIs. The task focuses on understanding API communication, testing API responses, identifying security risks, and improving cybersecurity awareness.

2. Objectives

- Understand API communication using HTTP requests and responses.
- Perform API testing using Postman.
- Analyze API responses and status codes.
- Identify common API security risks.
- Document findings and recommendations.

3. API Testing Overview

API Endpoint	Method	Status
https://jsonplaceholder.typicode.com/users	GET	200 OK
https://jsonplaceholder.typicode.com/posts	GET	200 OK

4. API 1 Analysis – Users API

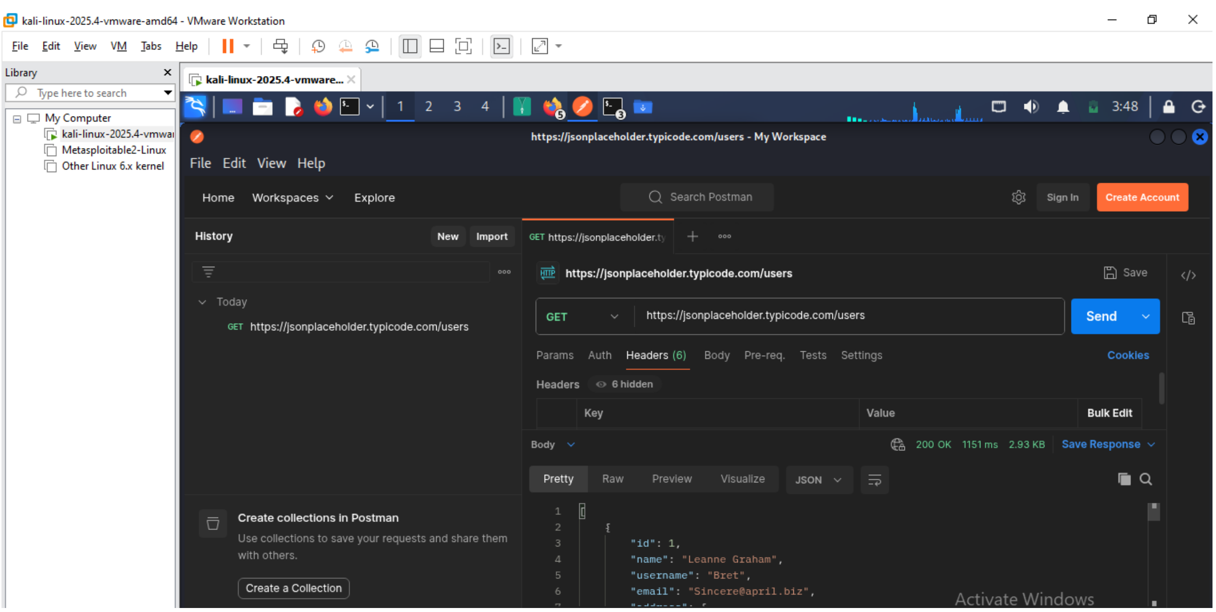

Fig1: API Testing using Postman – Users API Response

Observation

The Users API returned JSON-formatted user data successfully. The API was publicly accessible without authentication or API keys.

Identified Risks

- No authentication required.
- Public data exposure.
- Open API access.
- No visible rate limiting.

5. API 2 Analysis – Posts API

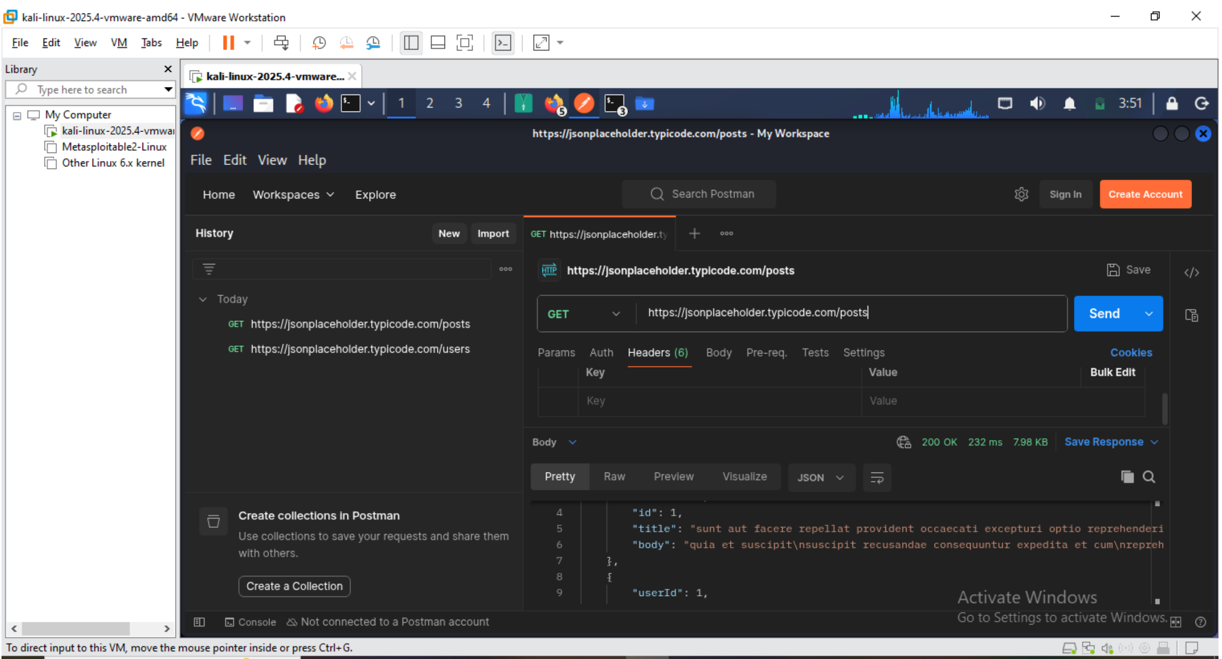

Fig 2: API Testing using Postman – Posts API Response

Observation

The Posts API returned post data successfully in JSON format. The endpoint was publicly accessible and allowed repeated requests.

Identified Risks

- No authentication required.
- Public data exposure.
- Open API access.
- No visible rate limiting.

6. Risk Analysis

Risk	Severity	Observation
No Authentication	Medium	API accessible publicly
Public Data Exposure	Medium	User data visible
Open API Access	Medium	No restrictions
No Rate Limiting	Low	Unlimited requests possible

7. Security Recommendations

- Implement API authentication.
- Restrict unauthorized public access.
- Enable rate limiting.
- Protect sensitive information.
- Use secure communication methods.

8. Learning Outcome

Through this task, I learned how APIs communicate using HTTP requests and responses, how to test APIs using Postman, and how to identify common API security risks.

9. Conclusion

The tested APIs responded successfully and returned JSON-formatted data using GET requests. Several API security concerns were identified, including lack of authentication and unrestricted public access.